

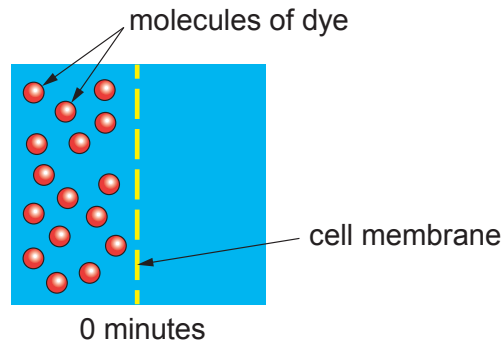
2. A student, Lisa, is studying how substances are transported into and out of cells.

(a) Oxygen enters the cell by diffusion. Define the process of diffusion. [1]

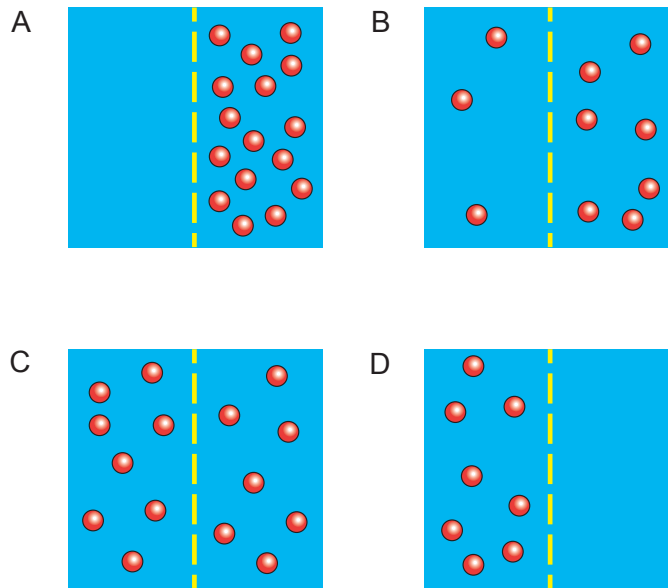
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(b) In her schoolbook, Lisa has diagrams of a model to show diffusion of dye into a cell. The dye takes 30 minutes to reach equilibrium.



Lisa thinks the diagram that would show the correct arrangement of dye molecules she would expect to see after 30 minutes is diagram A.



Explain whether you agree with Lisa. [3]

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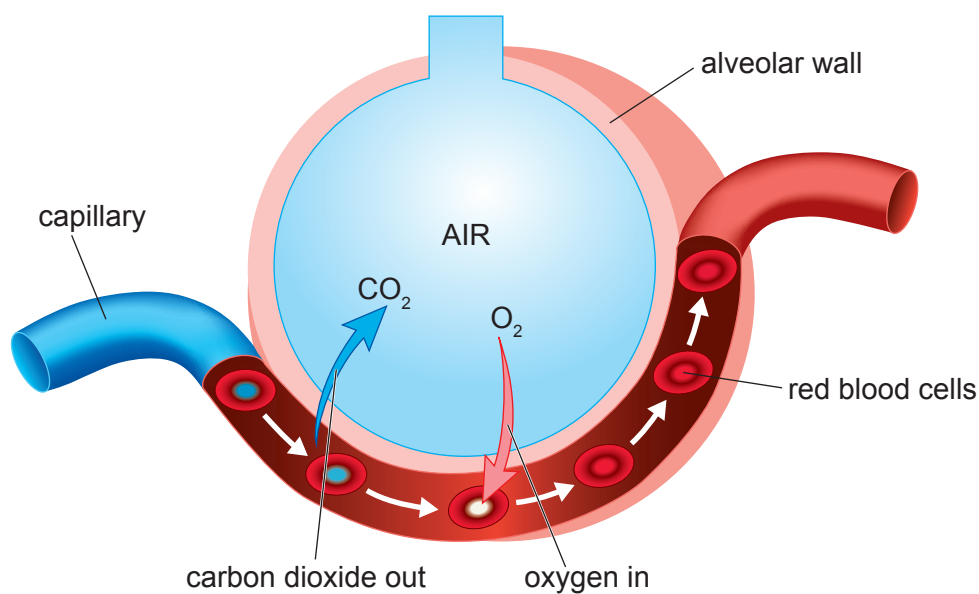
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- (c) Oxygen and carbon dioxide diffuse into and out of blood surrounding the alveoli.



Use the diagram to explain how diffusion is always maintained.

[3]

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- (d) Describe **two** differences between diffusion and active transport.

[2]

1.

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2.

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